

ENSF 380

TOPIC 07: PRIMITIVE DATA WRAPPERS

Wrapper Classes

Primitive data types have class versions:

- Boolean
- Character
- Integer
- Long
- Float
- Double
- Short
- Byte

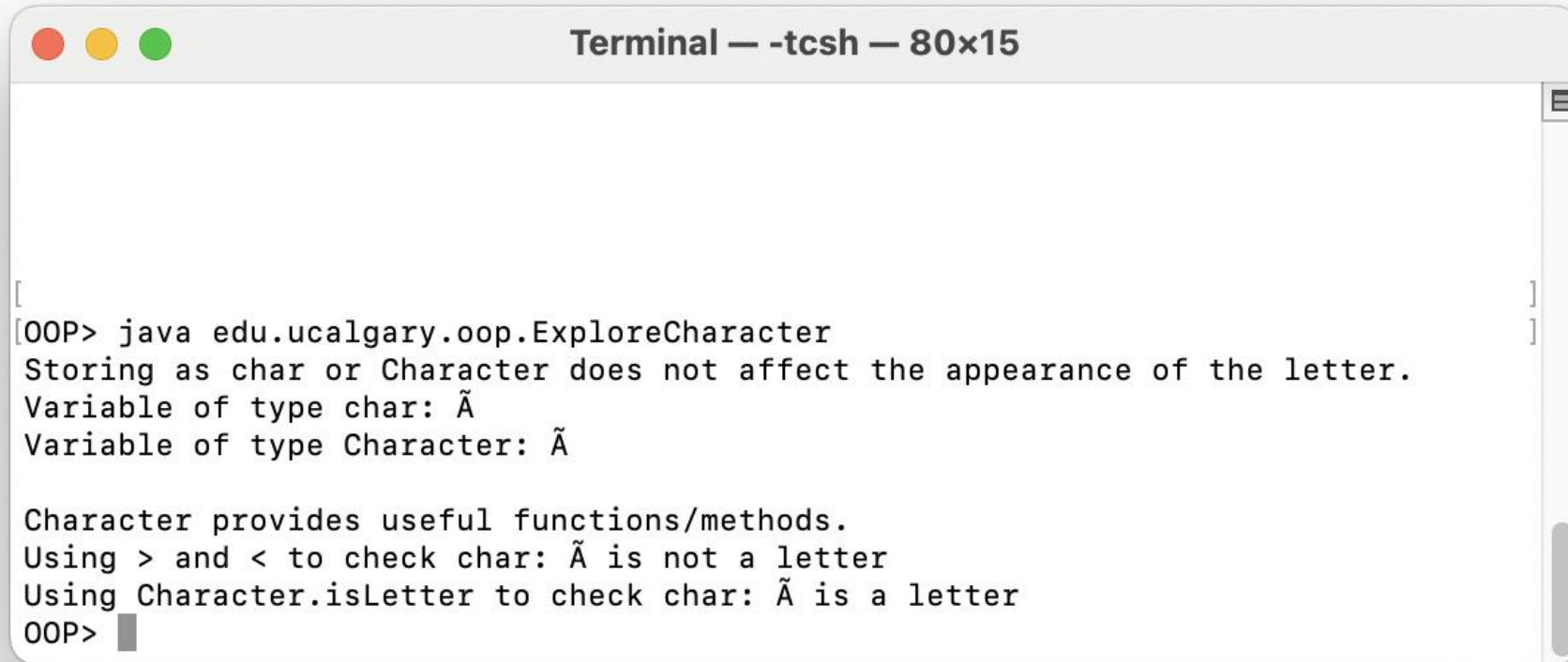
```
...
public static void main(String[] args) {
    char justChar = ('\u00C3');
    Character characterTest = Character.valueOf(justChar);

    // They can be used interchangeably
    System.out.println("Storing as char or Character does not affect" +
        " the appearance of the letter.");
    System.out.println("Variable of type char: " + justChar);
    System.out.println("Variable of type Character: " + characterTest);
    System.out.println();
}
...
```

```
...  
  
// Naïve approach to checking characters  
System.out.print("Using > and < to check char: ");  
if ((justChar >= 'a' && justChar <= 'z') ||  
    (justChar >= 'A' && justChar <= 'Z')) {  
    System.out.println(justChar + " is a letter");  
} else {  
    System.out.println(justChar + " is not a letter");  
}  
  
// Using a method of the Character class  
    System.out.print("Using Character.isLetter to check char: ");  
if (Character.isLetter(justChar)) {  
    System.out.println(justChar + " is a letter");  
} else {  
    System.out.println(justChar + " is not a letter");  
}  
...  

```

Character



```
Terminal — -tcsh — 80x15

[
[OOP> java edu.ucalgary.oop.ExploreCharacter
Storing as char or Character does not affect the appearance of the letter.
Variable of type char: Ã
Variable of type Character: Ã

Character provides useful functions/methods.
Using > and < to check char: Ã is not a letter
Using Character.isLetter to check char: Ã is a letter
OOP> █
```

When to Use

Advantages to Primitives:

- Objects use more memory

Advantages of Wrappers:

- Perform activities restricted to objects, such as in collections and generics

Remember, functions provided by wrapping classes can be used without instantiating an object.

Summary: Knowledge

Methods provided by class wrappers

- Boolean
- Character
- Integer
- Long
- Float
- Double
- Short
- Byte

Additional Information

Related reading

- Volume 1, Chapter 5.4: Object Wrappers and Autoboxing